

MICHAELA BREZINOVA

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EDUCATION

- **UNIVERSITY OF CAMBRIDGE**, PhD in Computational Chemistry (Vendruscolo Lab) **Oct 2023 - Present**
 - Developing machine learning-driven computational pipelines for drug discovery, primarily in neurodegenerative disorders, with a focus on challenging and historically undruggable targets.
 - **Active learning & virtual screening:** Extended and adapted active learning-based Deep Docking framework (Gentile et al., Nature Protocols 2022) for ML-driven discovery of binders to A β 42 disordered monomers, A β 42 fibrils, and tau fibrils, and for designing subunit-selective ligands for GABAA receptors
 - **First-in-class binder discovery:** Identified the first reported small-molecule inhibitors of A β 42 fibril secondary nucleation, a key mechanism in Alzheimer's disease progression, achieving low-nanomolar affinities and a 54% experimental hit rate using Deep Docking against an ultra-large chemical library (Brezinova et al., Molecular Systems Biology 2025, see *Publications*)
 - **Covalent inhibitor discovery:** Developed a hierarchical large-library screening pipeline combining rapid sequence-based prescreening with high-accuracy structural refinement for covalent inhibitor discovery (validation pending)
 - **Protein-protein interaction disruption:** Curated PPI disruptors dataset and designing a multi-task learning pipeline for virtual screening of protein-protein interaction disruptors leveraging protein language model embeddings, trained on heterogeneous bioactivity data with varying mechanistic annotation depth (in development)
 - **Other:**
 - **Protein phase separation tool:** Adapted the DropFit method (Amico et al., eLife 2024) into a published web server for calculating critical concentration in protein liquid-liquid phase separation (Brezinova et al., Journal of Molecular Biology, 2025, see *Publications*)
 - Large-scale molecular dynamics simulations of nanobodies
 - **Technical Skills and Tools:** *Python, PyTorch, HPC (Slurm workload manager), TensorFlow, AlphaFold3, Boltz-2, Bash, AutoDock Vina, OpenEye Suite, RDKit, Open Babel, Molecular Dynamics (GROMACS)*
- **UNIVERSITY OF CAMBRIDGE**, MPhil in Computational Biology **Oct 2021 - Nov 2022**
 - Grade: **Distinction**
- **UNIVERSITY OF EDINBURGH**, BSc Honours in Mathematics and Computer Science **Sep 2016 - Jun 2020**
 - Grade: **First Class**

WORK EXPERIENCE

- **X, The Moonshot Factory (formerly Google X)**, PhD Resident **(Mountain View, USA) Jul 2026 - Present**
 - Early-stage investigations with a focus on AI lab automation
- **UNIVERSITY OF CAMBRIDGE**, Research Assistant **(Cambridge, UK) May 2023 - Oct 2023**
 - Applied unsupervised machine learning to analyze large alpha-synuclein aggregate pathology in DAB-stained microscopy images of Parkinson's patient tissue, as part of the global Aligning Science Across Parkinson's (ASAP) initiative
 - Developed a kinetic model of aggregate growth from image-derived data.
 - **Technical Skills and Tools:** *Python, TensorFlow, FIJI (ImageJ)*
- **ILLUMINA**, Bioinformatics Intern **(Cambridge, UK) Aug 2022 - Nov 2022**
 - Developed a method for detecting variants in nucleotide sequences based on haplotype diversity - resulting in a granted patent (Andrews & Brezinova, 2024).
 - **Technical Skills and Tools:** *R, Nextflow, Make, Bash, BCFTools*
- **GOOGLE**, Software Engineer at Google Health **(London, UK) Sep 2020 - Oct 2021**
 - Designed, implemented and delivered the first version of a critical front-end latency feature, important for meeting the release performance requirements for Care Studio, a clinical platform unifying healthcare data.
 - Delivered full-stack features optimizing data presentation for clinicians, collaborating closely with clinicians, product managers, and UX designers.
 - **Technical Skills and Tools:** *TypeScript, Java, C++, Soy (Closure templates)*
- **CERN**, Remote Summer Collaboration **(Remote) Jun 2020 - Aug 2020**
 - Designed and delivered a new profile customisation for ALICE (A Large Ion Collider Experiment) O2 InfoLogger, built for long-term durability across the 10+ year duration of the experiment. (<https://cds.cern.ch/record/2728718>)

- **Technical Skills and Tools:** *NodeJS, SQLite, HyperScript, JavaScript*

- **GOOGLE**, SWE Intern at Google Travel NBU (Next Billion Users) (Zurich, Switzerland) Jul 2019 - Sep 2019
 - Built a flexible dates feature for a train booking platform integrated within Google Pay for the Indian market
 - **Technical Skills and Tools:** *TypeScript, Java, Soy (Closure templates)*
- **GOOGLE**, STEP Intern at Google Account (Munich, Germany) Jul 2018 - Sep 2018
 - Worked on the redesign of the Takeout page (takeout.google.com). The new design shipped to production and remains live.
 - **Technical Skills and Tools:** *TypeScript, Java, Soy (Closure templates)*

PUBLICATIONS

- Hanke, J., Pujalte Ojeda, S., Cheong, R.W., Glasstetter, L.M., Baker, E., Lam, H.Y.I., **Brezinova, M.**, Louet, A.A.B., Zhang, S. and Vendruscolo, M., 2026. OmniBind: Proteome-Wide Promiscuity Predictions for Early-Stage Drug Screening. *bioRxiv*, pp.2026-03.
- Lam, H.Y.I., Pujalte Ojeda, S., **Brezinova, M.**, Hanke, J., Ong, X.E., Mu, Y. & Vendruscolo, M., 2026. Metadiffusion: inference-time meta-energy biasing of biomolecular diffusion models. *bioRxiv*, p.2026.02.10.704873. <https://doi.org/10.64898/2026.02.10.704873>
- **Brezinova, M.**, Brotzakis, Z.F., Horne, R.I., Roy Chowdhury, V., Gregory, R.C., Bian, Y., González Díaz, A., Gentile, F. and Vendruscolo, M., 2025. Identification of potent high-affinity secondary nucleation inhibitors of Aβ42 aggregation from an ultra-large chemical library using deep docking. *EMBO Molecular Systems Biology*, pp.1-19. <https://doi.org/10.1038/s44320-025-00159-5> (GitHub link)
- **Brezinova, M.**, Fuxreiter, M. and Vendruscolo, M., 2025. DropFit: Determination of the Critical Concentration for Protein Liquid-Liquid Phase Separation. *Journal of Molecular Biology*, p.169294. <https://doi.org/10.1016/j.jmb.2025.169294> (webserver link, GitHub link)
- Amico, T., Dada, S., Lazzari, A., **Brezinova, M.**, Trovato, A., Vendruscolo, M., Fuxreiter, M., & Maritan, A. (2024) 'A scale-invariant log-normal droplet size distribution below the critical concentration for protein phase separation', *eLife*, 13, RP94214. <https://doi.org/10.7554/eLife.94214.2>

PATENTS

- Andrews, D.J., **Brezinova, M.**, 2024. Detecting variants in nucleotide sequences based on haplotype diversity (<https://patents.google.com/patent/WO2025090883A1/>)

TALKS

- Talk at In Silico #001 event series (<https://luma.com/dw9ijcrm>) (London, UK) Jul 2025
- Short Talk at AI in Drug Discovery at **IRB Barcelona Biomedicine Conference** (Barcelona, Spain) Apr 2025
- Talk at Bio2Brain Network's (<https://bio2brain.eu/>) AI Workshop (Online) Jun 2024

SKILLS

- **Programming Languages:** *Python (experienced), TypeScript (experienced), Bash (experienced), Soy (Closure Templates) (experienced), HTML (experienced), CSS (experienced), Java (skilful), R (skilful), C++ (skilful), Haskell (familiar)*
- **Cheminformatics:** *AutoDock Vina (experienced), OpenEye Suite (experienced), RDKit (experienced), Open Babel (Obabel) (experienced), AlphaFold3 (experienced), Boltz-2 (skilful), Molecular Dynamics(GROMACS) (familiar)*
- **Machine Learning:** *PyTorch (skilful), TensorFlow (skilful)*
- **Computational Tools and Frameworks:** *Slurm (experienced), Nextflow (skilful), Make (skilful)*
- **Spoken languages:** *Slovak (native proficiency), Czech (native proficiency), English (full professional proficiency), German (limited working proficiency), Mandarin Chinese (elementary proficiency)*

VOLUNTEERING AND LEADERSHIP - ACADEMIC

- Chemistry Lab Demonstrator at the University of Cambridge (Cambridge, UK) Jan 2024 - Feb 2025
- Code First: Girls, Introduction to Web Development course instructor (Edinburgh, UK) Jan 2020 - Mar 2020
- Director of Media for the Women in Tech Conference in Edinburgh (Edinburgh, UK) May 2017 - Jun 2018
- MathPALS Student Leader at the University of Edinburgh (Edinburgh, UK) Sep 2017 - Dec 2017

COMPETITION ACHIEVEMENTS

- 3rd place at International Team High School Internet Mathematical Olympiad
- 3rd place at National Scientific Research Paper's Competition for high school students in Mathematics and Physics category with paper on topic *Mathematics and Music*